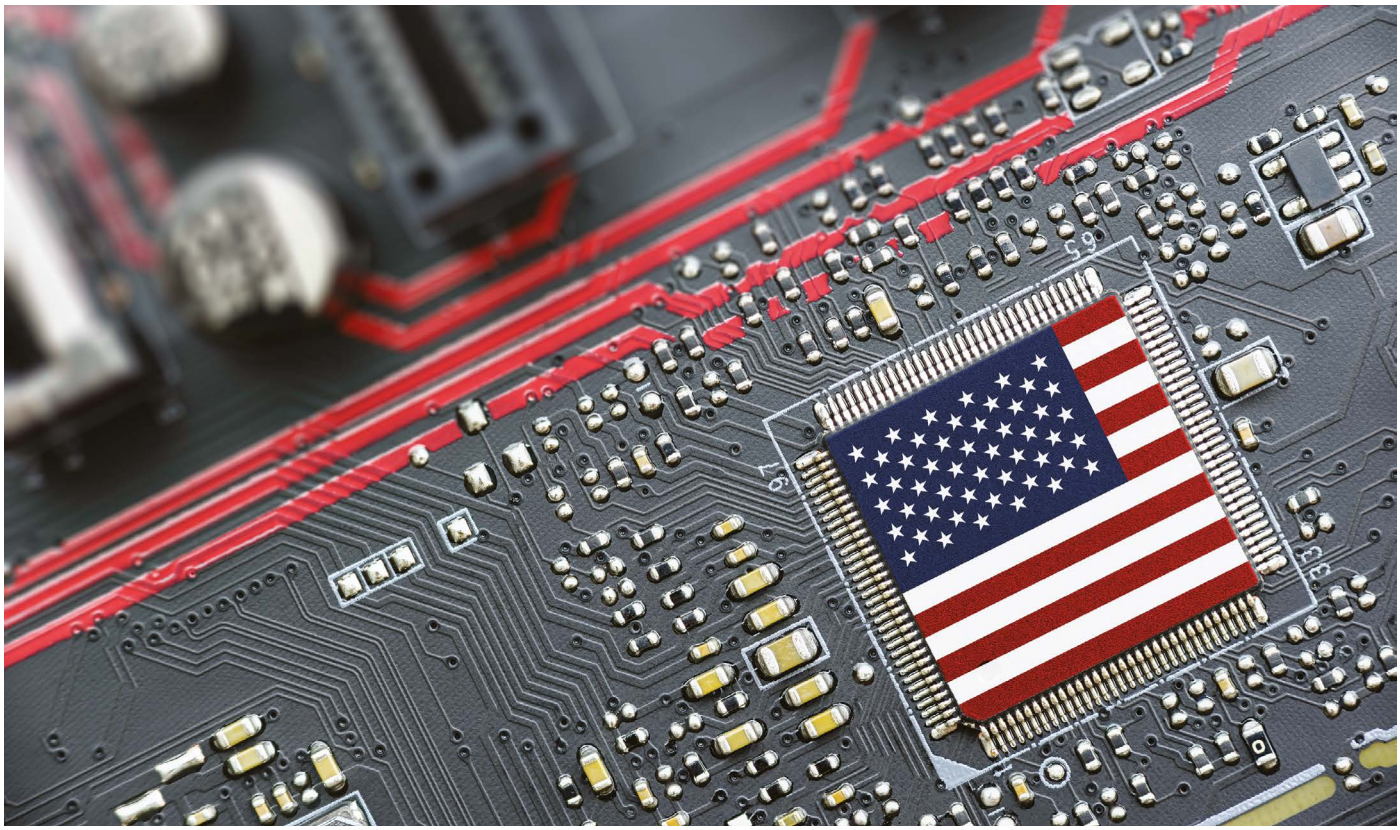
News in focus



WILLIAM POTTER/ISTOCK VIA GETTY

The Genesis Mission aims to provide researchers with the computing power and data sets necessary to train artificial-intelligence models.

TRUMP'S AI 'GENESIS MISSION': WHAT ARE THE RISKS AND OPPORTUNITIES?

National laboratories have been told to broaden access to their data sets to accelerate research as part of the federal government's AI platform. But who stands to benefit?

By Elizabeth Gibney, Alexandra Witze & Jenna Ahart

The White House has launched a plan to accelerate research in the United States, by building artificial intelligence (AI) models on the rich scientific data sets held by the country's 17 national laboratories, as well as harnessing their enormous computing resources.

An executive order issued on 24 November instructs the US Department of Energy (DoE) to create a platform through which academic researchers and AI firms can create powerful AI models using the government's scientific data. Framed as part of a race for global technology dominance, it lists collaborations with technology firms including Microsoft, IBM, OpenAI, Google and Anthropic, as well as quantum-computing companies such as Quantinuum. Such a vast public–private partnership would give companies unprecedented access to federal scientific data sets for AI-driven analysis.

The effort, dubbed the Genesis Mission, aims to "double the productivity and impact of American research and innovation within a decade", in a variety of fields from fusion energy to medicine. The project expects to "unlock breakthroughs in medicine, energy, materials science and beyond", says Michael Kratsios, the US president's science adviser. It also aims to build AI agents – general models with the ability to harness tools such as specialized software and coding suites – that can generate hypotheses and automate research workflows.

Labs around the world are already training AI systems on scientific data, to boost their capabilities in scientific domains and attempting to use AI models to make discoveries. But some researchers remain sceptical that general AI tools are capable of generating truly fresh insights, and warn that their inherent flaws make the value of agents unclear.

The new US initiative formalizes and expands ongoing AI research efforts by the administration of President Donald Trump.

"The impact is that it enables many more scientists and researchers to have access to all of the infrastructure that they need to explore important scientific questions that the country cares about," says Lynne Parker, a robotics engineer at the University of Tennessee, Knoxville, who led AI-policy initiatives for the administrations of Trump and his predecessor, Joe Biden, but was not involved in the current initiative. "That really has not been possible before."

Trump's team has been working to funnel money and attention to AI projects even as it tries to gut federal research spending more broadly. The White House has the power to shape the direction of research at the DoE's network of national laboratories. It did not give an estimated price tag for the AI initiative; any extra funding beyond the laboratories' normal budgets would have to be approved by the US Congress.

Nature looks at how the project might affect researchers and AI companies, and its promises and risks.

What are government-funded scientists being asked to do?

The scale and timeline of the plan is ambitious. In 60 days, the DoE is expected to create a list of 20 potential science and technology challenges for the project to tackle, in areas of national priorities such as nuclear fusion, quantum information science and crucial materials. The agency is supposed to create a full inventory of available federal computing resources and identify initial data assets to use, then work out how to safely include external data sets. The administration expects to demonstrate the platform's capability for one of these research challenges in nine months.

These early steps will probably build on projects that are already under way at the national laboratories. For instance, Oak Ridge National Laboratory in Tennessee has been working on advancing AI research using a hybrid approach that uses both quantum and classical computing. Lawrence Berkeley National Laboratory in California is using AI to find ways to speed up network traffic.

The "DoE has been making a case for 'AI for science' for over seven years, and this executive order is the starting pistol to get on with it", says Michael Norman, an astrophysicist at the University of California, San Diego, and former director of the San Diego Supercomputer Center. "It is an exciting direction indeed."

What are companies being asked to do?

The project has named more than 50 collaborating companies, including some that have already been working on their own 'AI scientists'. FutureHouse, a start-up based in San Francisco, California, for instance, launched a commercially available, AI-driven research platform last month.

The precise role of these private companies in the Genesis plan remains unclear — although Trump's executive order says the project will entail "collaboration with external partners possessing advanced AI, data, or computing capabilities or scientific domain expertise". Such partnerships could include research agreements to jointly develop technologies, or a user-facility agreement for external researchers to conduct work in government facilities. Chip-manufacturing and computer companies such as NVIDIA, Advanced Micro Devices and Hewlett Packard Enterprise have reportedly agreed to build facilities in national labs, according to a report by *The New York Times* (see go.nature.com/4wvw2rh).

"Such data sets can immediately accelerate the development of trustworthy, science-capable AI models."

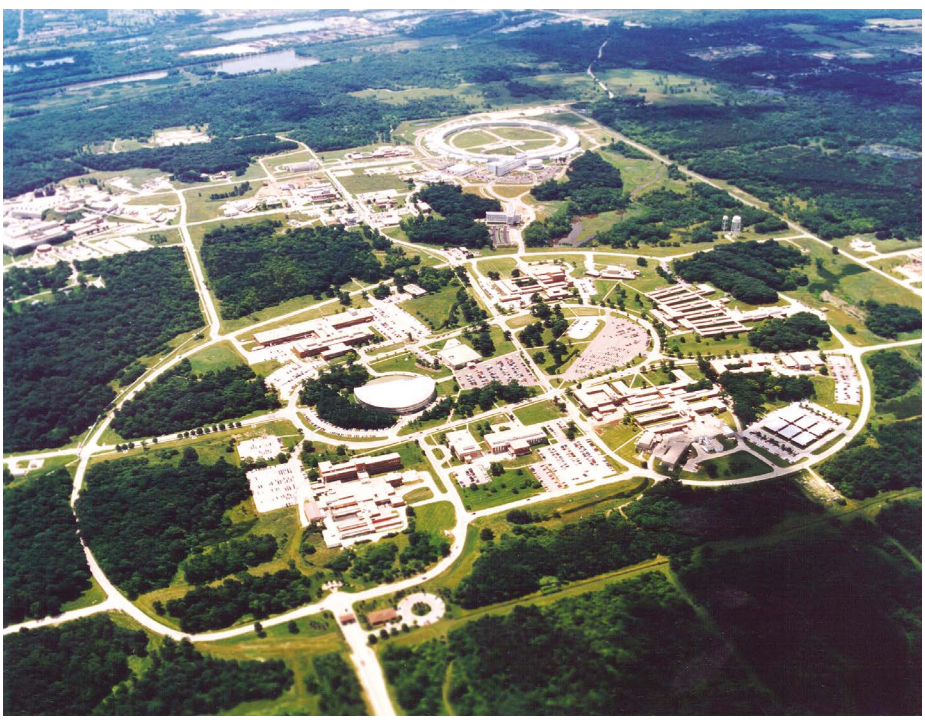
Some partnerships are already under way. In October, for instance, Argonne National Laboratory in Illinois announced a partnership with NVIDIA and technology firm Oracle to build two next-generation AI supercomputing systems. At least two national laboratories already have arrangements with the company OpenAI, based in San Francisco, California, to host local AI models that can process classified data on computers in the facilities. In February, the company held an 'AI jam session' with researchers from nine US national labs that allowed scientists to test the use of OpenAI's reasoning models in their specific domains.

What are the opportunities?

The new Genesis Mission is meant to provide "secure access to appropriate datasets, including proprietary, federally curated, and open scientific datasets, in addition to synthetic data generated through DOE computing resources", according to the executive order. Creating a national-scale platform that harnesses rich data sets usually housed in the national laboratories could be a boon for researchers. Although task-specific models such as the protein-folding model AlphaFold were built on open scientific data sets, general-purpose AI systems such as OpenAI's GPT-5 are thought to be largely built on data scraped from the Internet.

It isn't yet clear which data sets would be involved, but possible candidates include information gathered at facilities such as the Spallation Neutron Source at Oak Ridge National Laboratory. That facility collects neutron-scattering data, which illuminate how materials behave at the atomic scale, says Georgia Tourassi, the lab's associate director for computing and computational sciences. "From atomistic measurements to exascale simulations, such data sets can immediately accelerate the development of trustworthy, science-capable AI models," she says.

Access to the DoE's computing power could



ARGONNE NATIONAL LABORATORY/GETTY

The Argonne National Laboratory in Illinois.

also accelerate research. Argonne's Aurora supercomputer, for example, uses graphics processing units, known as GPUs, that make it particularly suitable for AI, which requires huge parallel processing power.

What are the risks and challenges?

For starters, Congress might not allocate enough money to the DoE to achieve its ambitious plans, which the Trump administration compares "in urgency and ambition" with the Manhattan Project, the secret multi-billion-dollar US government programme that produced the first nuclear weapons. Trump has proposed cutting the DoE's science budget by 14% for the 2026 fiscal year and funding for AI might entail drawing funds from elsewhere in the budget.

Data security is another big question. Trump's executive order says all data will be handled consistently with regard to law, classification, privacy and intellectual property protections. Tourassi says she expects data to be made available "in alignment with the established data-sharing policies of our user facilities and sponsor programmes".

Then there's the issue of how to wrangle all these disparate data sets into a single user-friendly platform. "Transforming isolated datasets into a unified engine for discovery" will be a major challenge, wrote Darío Gil, the DoE under-secretary for science who will head the project, and physicist Kathryn Moler at Stanford University in California, in an editorial in *Science* (D. Gil and K. A. Moler *Science* **390**, 965; 2025). They note that some of the most successful big-science projects in recent years, such as the Protein Data Bank repository and the Large Hadron Collider particle accelerator at CERN, Europe's particle-physics laboratory near Geneva, Switzerland, succeeded because they were open source and relied on well-structured data — which is not always the case with large data sets.

Parker says she would have liked to have seen more in the Genesis Mission plans about how more US scientists could participate in this research. As it stands, most of the benefit might flow to researchers who already work with DoE-funded platforms.

The plan is also forging ahead without any comprehensive federal legislation to regulate AI. In January, Trump revoked an executive order that was created by Biden aimed at ensuring AI safety. The Trump administration has positioned itself pro-industry and called for federal funding for AI to be withheld from any state with "burdensome AI regulations".

Many researchers argue that AI applied to science can benefit humanity. But they say that companies pursuing intelligence with broad capabilities, beyond those of humans, and without strict safety guard-rails or limits on AI power, bring a range of risks.

KELLY VANDELLEN/ISTOCK VIA GETTY

The heat-tolerant amoeba was found at Lassen Volcanic National Park in California.

'FIRE AMOEBA' SURVIVES TEMPERATURES THAT KILL OTHER COMPLEX LIFE

The single-celled organism can grow at 63 °C, a record for eukaryotic life.

By Ewen Callaway

This cell likes it hot. A tiny, single-celled amoeba that can thrive at temperatures that kill all other known complex life — organisms whose cells contain a nucleus and internal structures — has been found.

The discovery questions the notion that such 'eukaryotic' life — which includes all animals and plants — is not suited to the kind of extreme conditions that can be tolerated by bacteria and other organisms lacking a cell nucleus.

"We need to rethink what's possible for a eukaryotic cell in a significant way," says Angela Oliverio, a microbiologist at Syracuse University in New York. The work, which has not yet been peer reviewed, was described in a preprint published on 24 November (H. B. Rappaport *et al*. Preprint at bioRxiv https://doi.org/qhcf; 2025).

Oliverio and fellow Syracuse microbiologist Beryl Rappaport were part of a team that discovered the organism at Lassen Volcanic National Park in northern California's Cascade mountain range. They named it *Incendiamoeba cascadensis*, which translates to 'fire amoeba from the cascades'.

The park is famous for gurgling acid lakes and incandescent geothermal pools, but *I. cascadensis* comes from a pH-neutral 'hot stream'. "It's the most uninteresting geothermal feature you'll find in Lassen," says Rappaport.

Water samples from the stream looked devoid of life under a microscope, but after culturing them with nutrients, the researchers spotted the amoeba growing at 57 °C, within the stream's temperature range. The scientists slowly raised the temperature, sailing past the previous eukaryote record of 60 °C. *I. cascadensis* was still able to divide at 63 °C and was still moving around at 64 °C. Even at 70 °C, the cells could form dormant 'cysts' that were capable of reactivating at cooler temperatures.

The hardiest bacteria and archaea can withstand much hotter temperatures: the archaean *Methanopyrus kandleri* holds the record for any life form at 122 °C. A few species of fungi and red algae held the previous record for eukaryotes; the upper limit for human and other mammalian cells seems to be around 43 °C.

The discovery of *I. cascadensis* underscores what can be gained by scouring the planet for new organisms, says Julia Van Etten, an